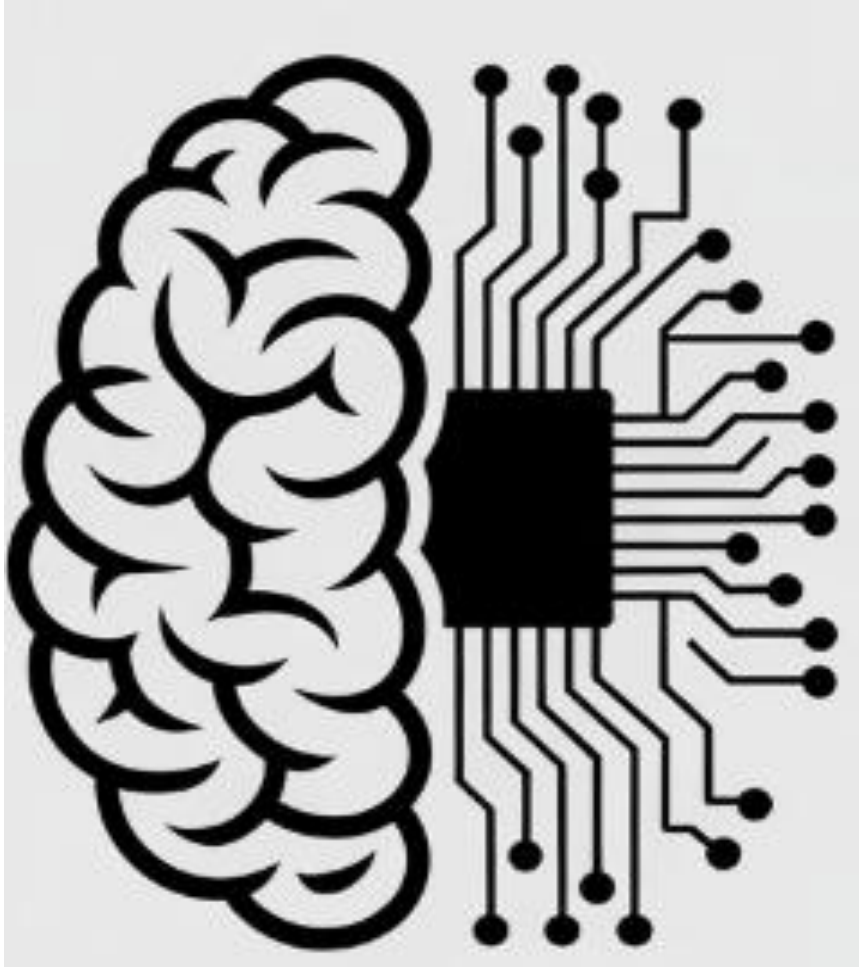




Regression

Dr. Sobia Tariq Javed

Supervised Learning

Training Data			
inputs			
$x_1 = [x_{11} \ x_{12} \ x_{13} \ x_{14}]$	targets / labels / ground truth $y_1 = 1$	predictions $\hat{y}_1 = 1$	
$x_2 = [x_{21} \ x_{22} \ x_{23} \ x_{24}]$	$y_2 = 2$	$\hat{y}_2 = 2$	
$x_3 = [x_{31} \ x_{32} \ x_{33} \ x_{34}]$	$y_3 = 1$	$\hat{y}_3 = 2$	
.			
.			
.			
$x_n = [x_{n1} \ x_{n2} \ x_{n3} \ x_{n4}]$	$y_n = 3$	$\hat{y}_n = 1$	

- We need to find a function that maps x and y for any of them.

$$\hat{y}_i = f(x_i; \theta)$$

Learnable
parameters

- How do we "learn" the parameters of this function?
- We choose ones that makes the following quantity small:

$$\sum_{i=1}^n \mathcal{L}(\hat{y}_i, y_i)$$

Loss function

Supervised Learning

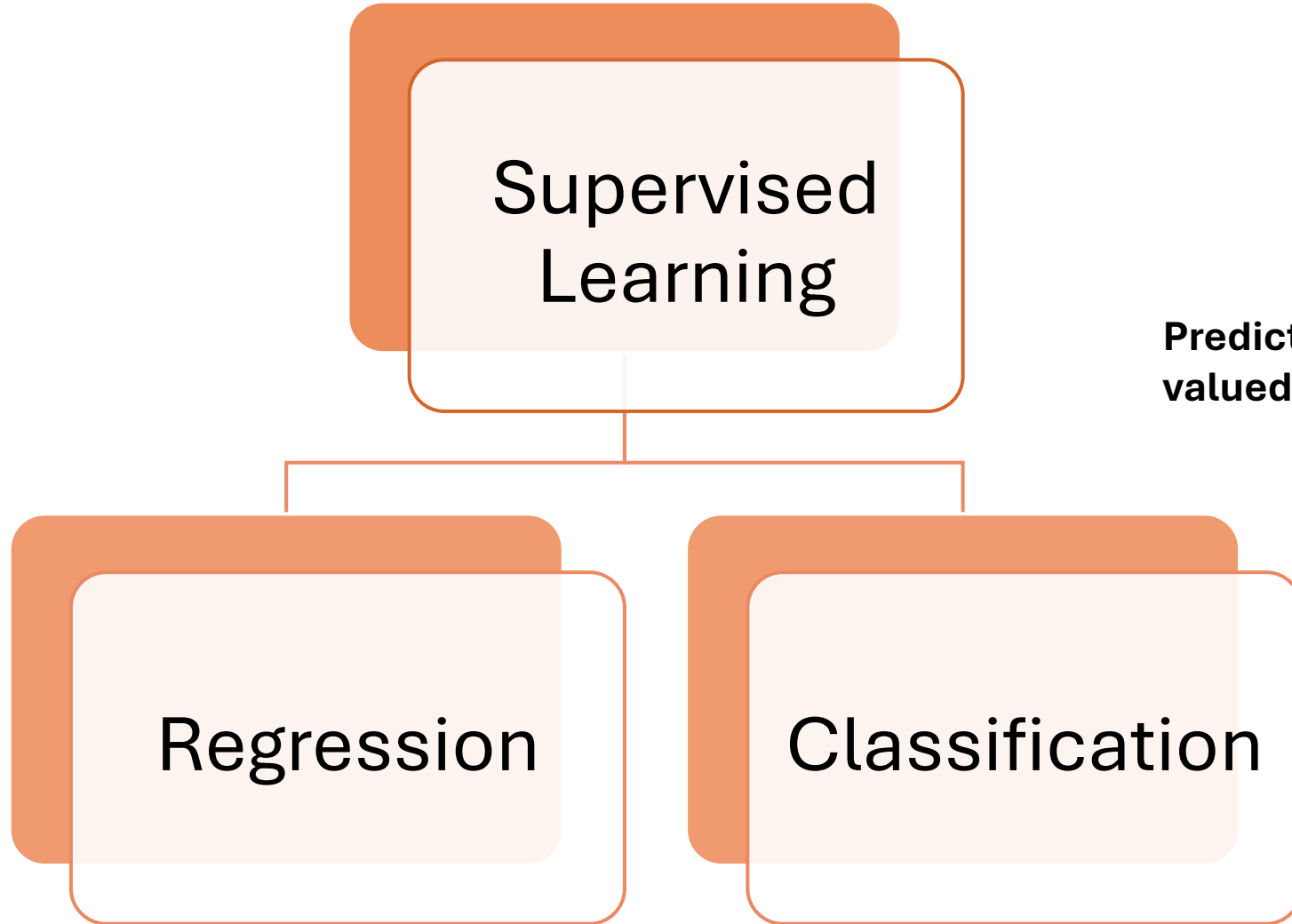
**Predict continuous
valued output (price)**

**Supervised
Learning**

**Predict discrete
valued output (0 or 1)**

Regression

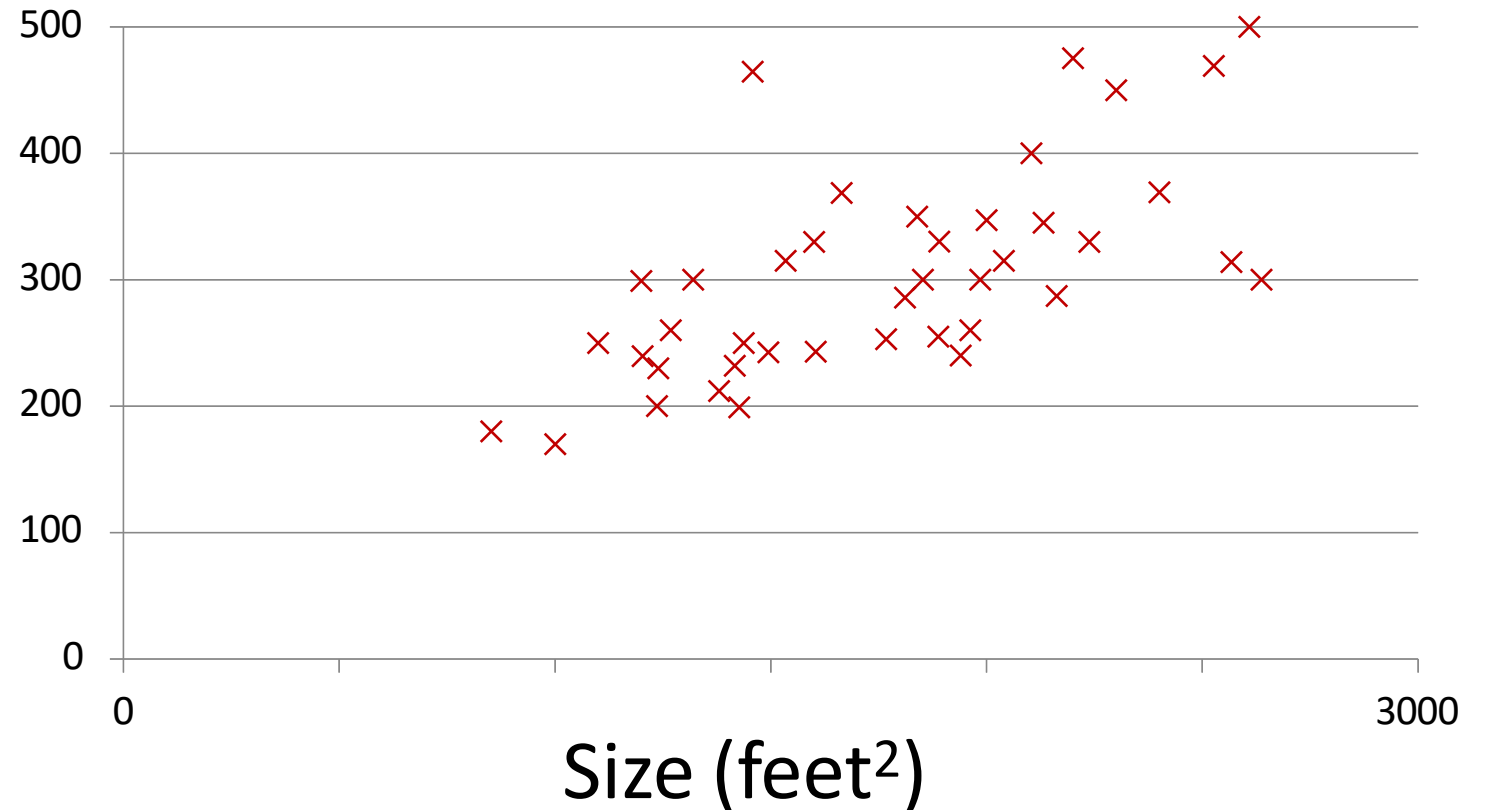
Classification



Regression

Housing Prices (Portland, OR)

Price
(in 1000s
of dollars)



Regression

Training set of housing prices	Size in feet ² (x)	Price (\$) in 1000's (y)
	2104	460
	1416	232
	1534	315
	852	178
	...	...

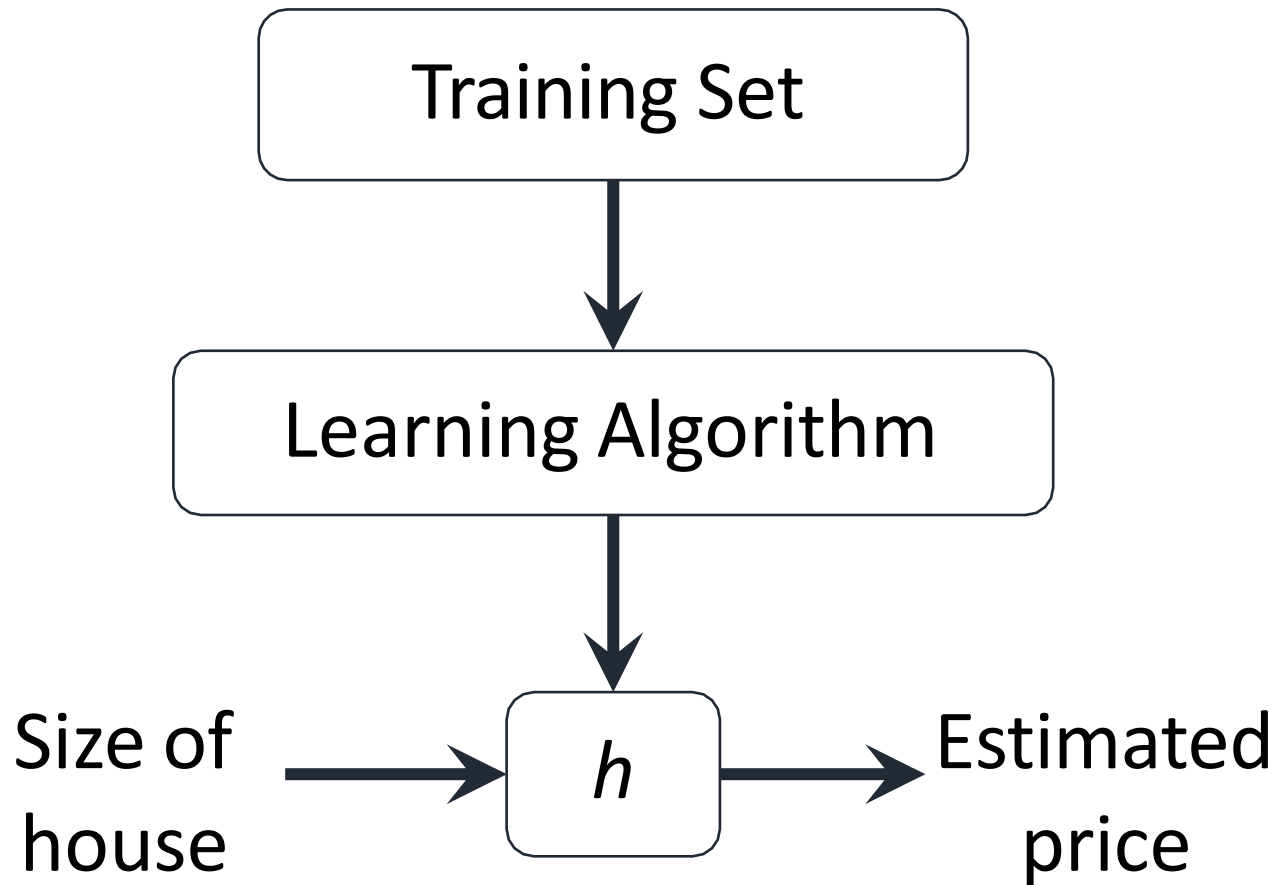
Notation:

m = Number of training examples

x's = “input” variable / features

y's = “output” variable / “target” variable

Regression



How do we represent h ?

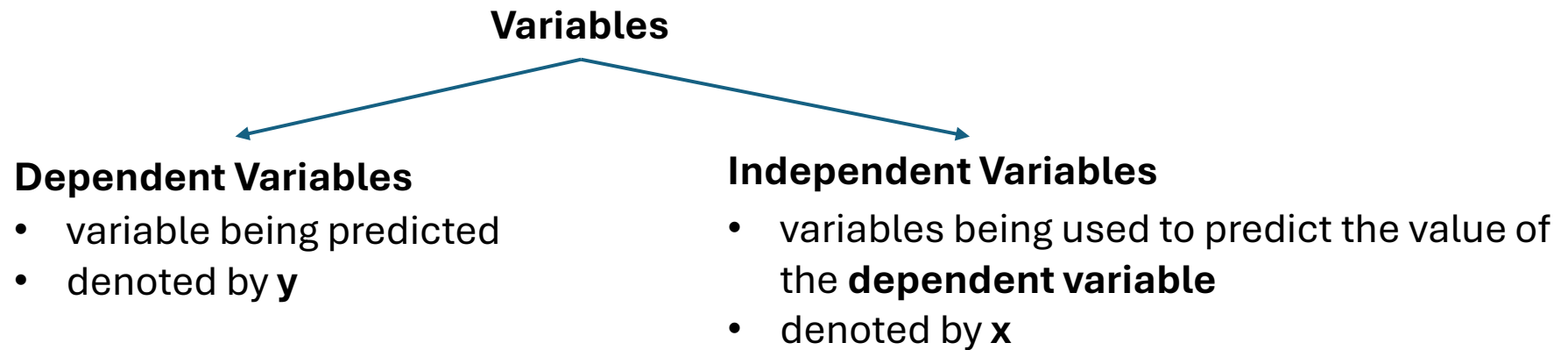
Linear regression with one variable.
Univariate linear regression.

Simple Linear Regression

- Simple Linear Regression Model
- Least Squares Method
- Co-efficient of determination
- Model Assumption

Simple Regression

- Managerial decisions often are based on the relationship between **two or more variables**.
- **Regression analysis** can be used to develop an equation showing how the variables are related.



Simple Regression

- Simple linear regression involves one **independent variable** and one **dependent variable**.
- The relationship between the two variables is approximated by a **straight line**.
- Regression analysis involving two or more independent variables is called **multiple regression**.

Simple Regression

- The equation that describes how **y** is related to **x** and an **error term** is called the **regression model**.
- The simple linear regression model is:

$$y = b_0 + b_1x + \varepsilon$$

- where:
 - b_0 and b_1 are called parameters of the model, ε is a random variable called the error term.

Simple Linear Regression Equation

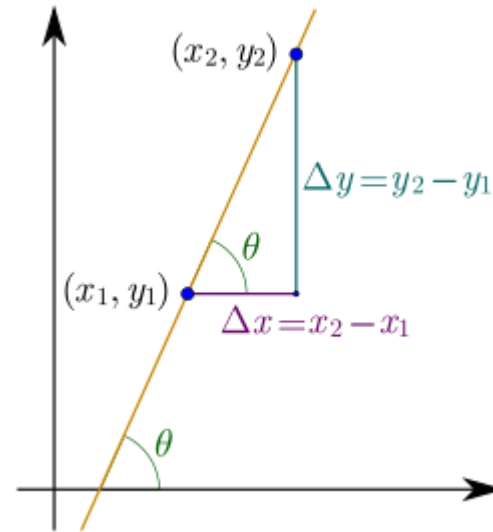
- The simple linear regression equation is:

$$\hat{y} = b_0 + b_1x$$

- Graph of the regression equation is a straight line.
- *Estimated value of $y = \hat{y}$*
- b_0 is the **y intercept** of the regression line.
- b_1 is the **slope** of the regression line.
- $\hat{y}$ is the expected value of y for a given x value.

Straight line equation

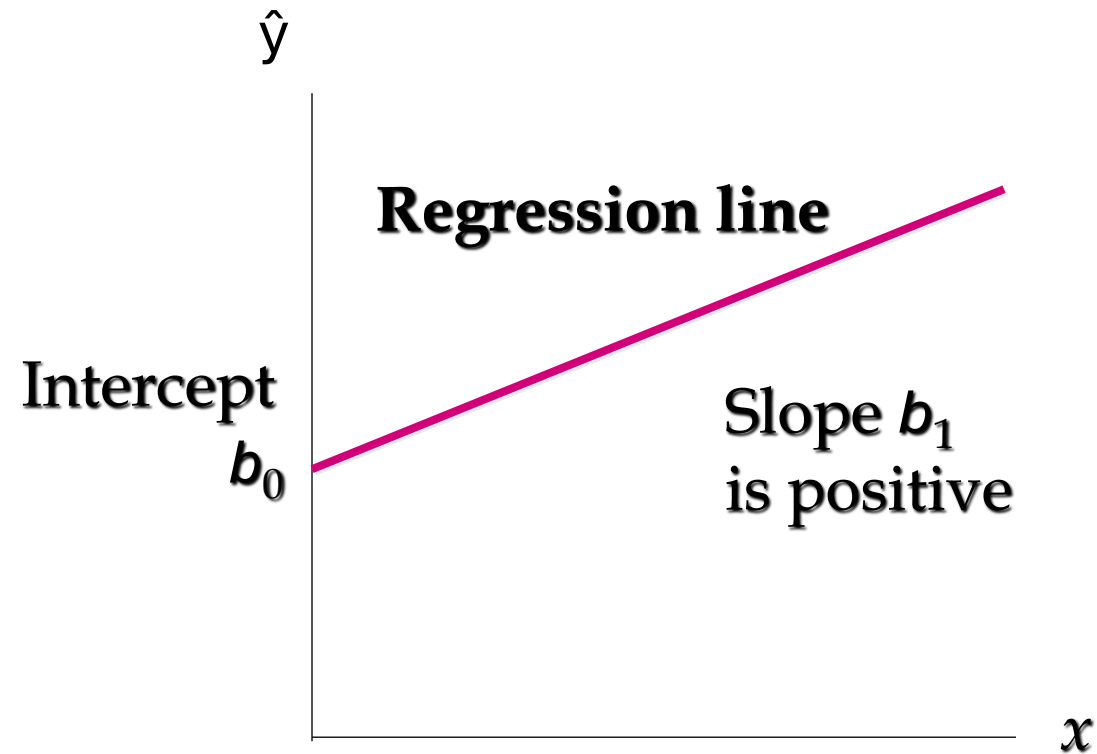
- $y = mx + c$



$$m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}.$$

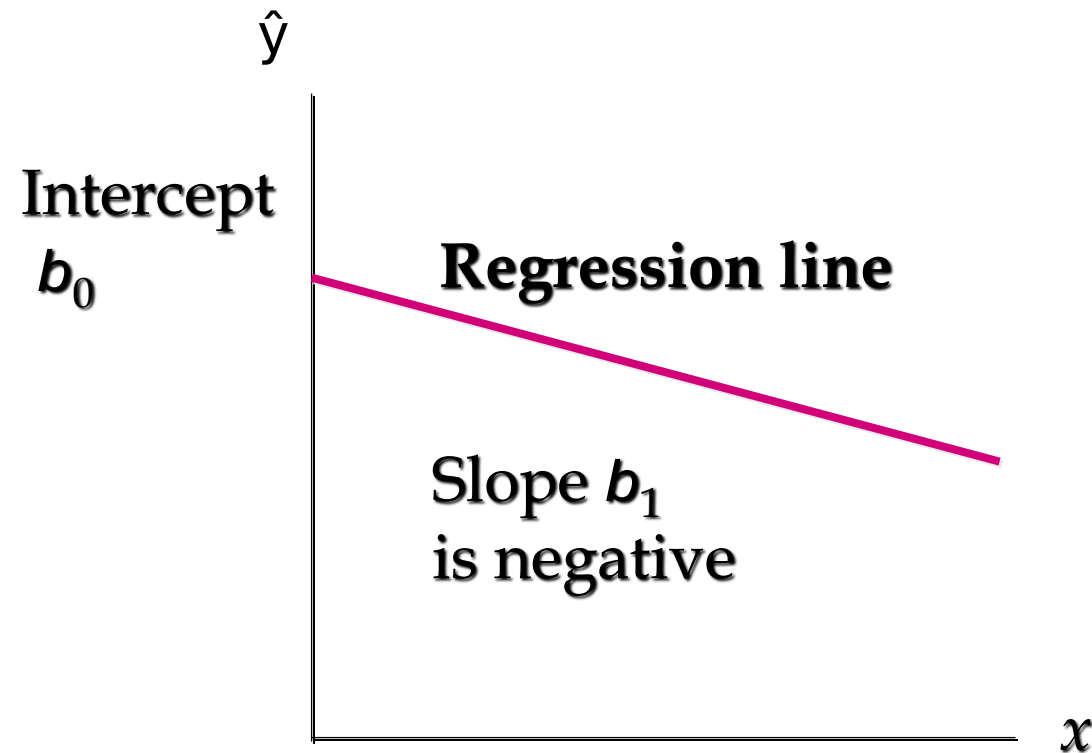
Simple Linear Regression Equation

- Positive Linear Relationship



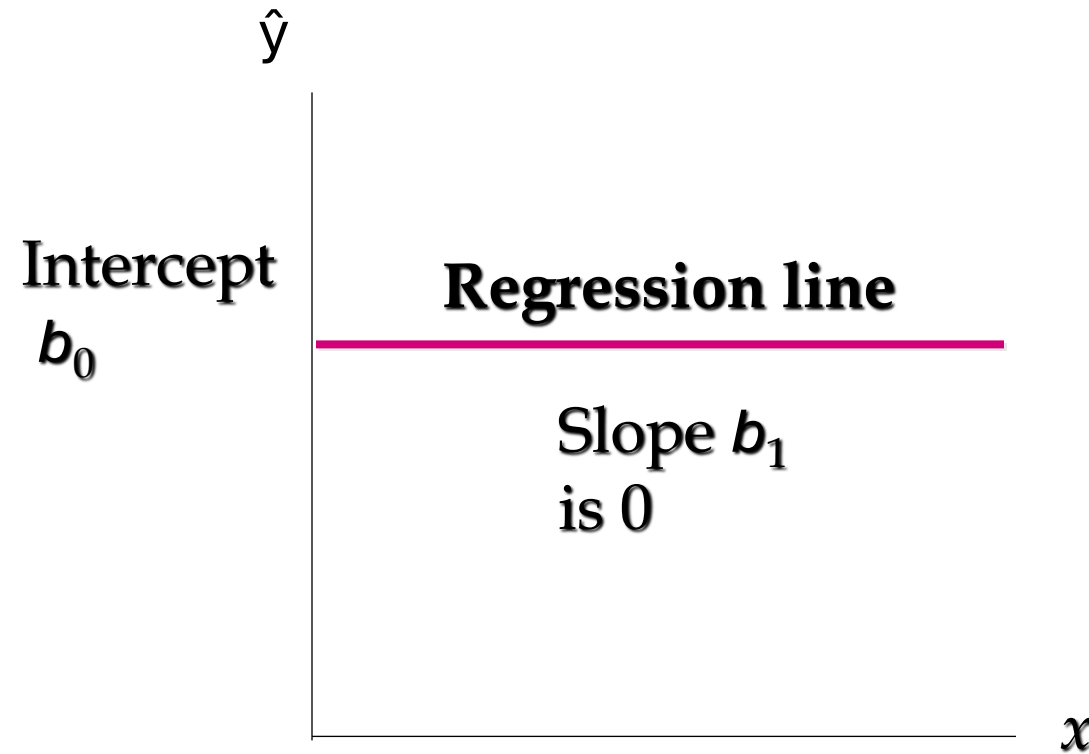
Simple Linear Regression Equation

- Negative Linear Relationship



Simple Linear Regression Equation

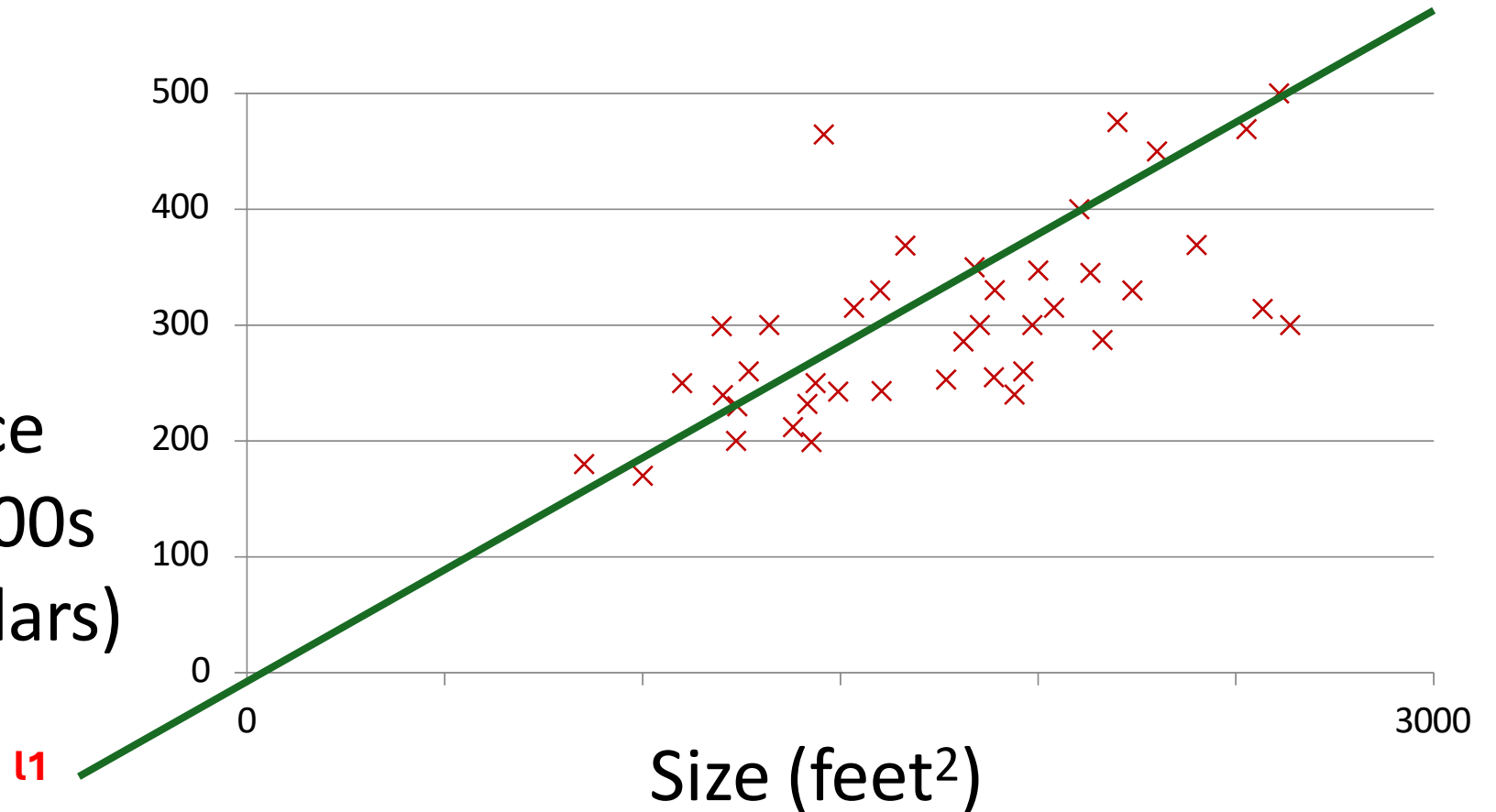
- No relationship



Regression

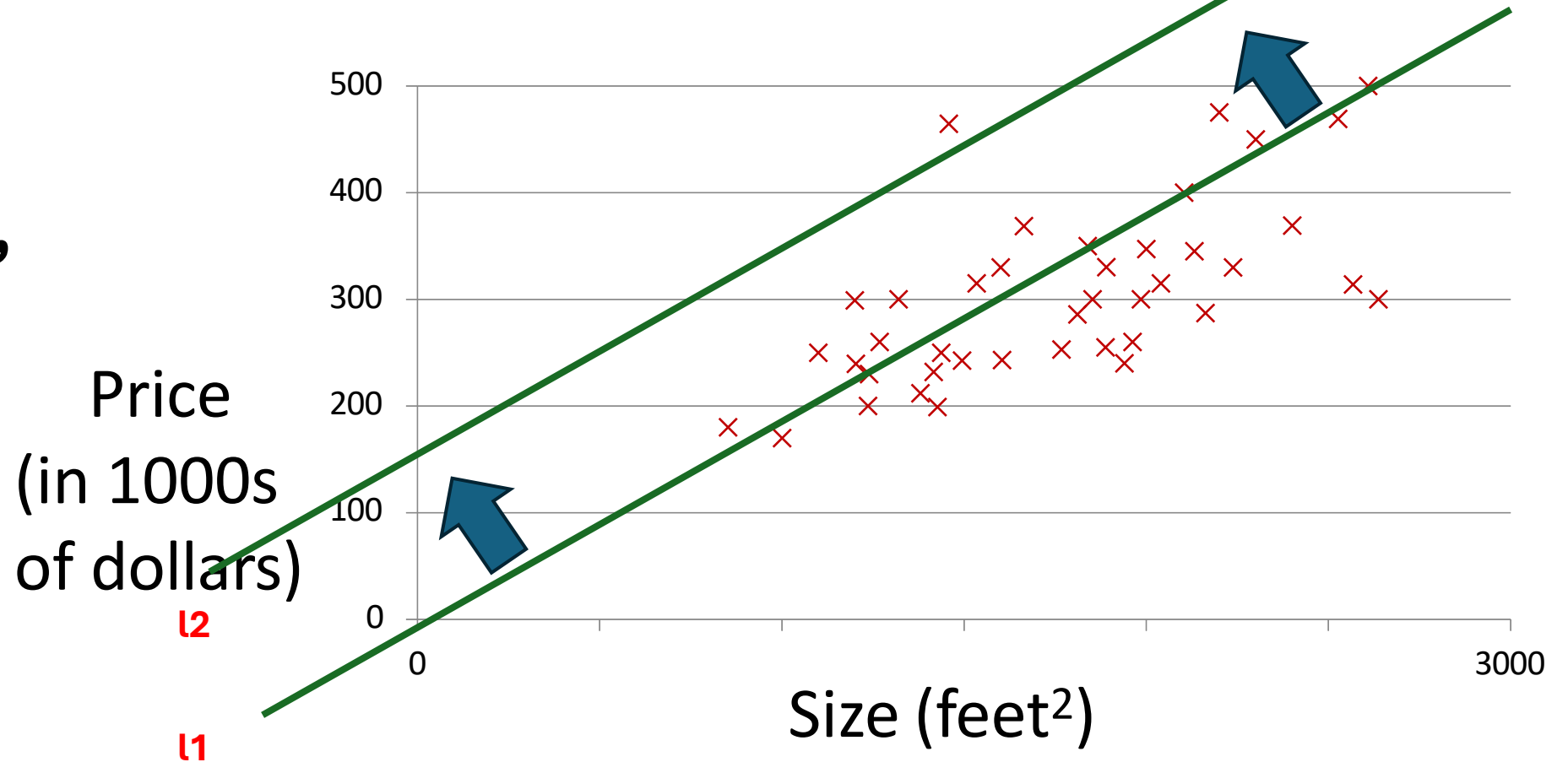
Housing Prices (Portland, OR)

Price
(in 1000s
of dollars)



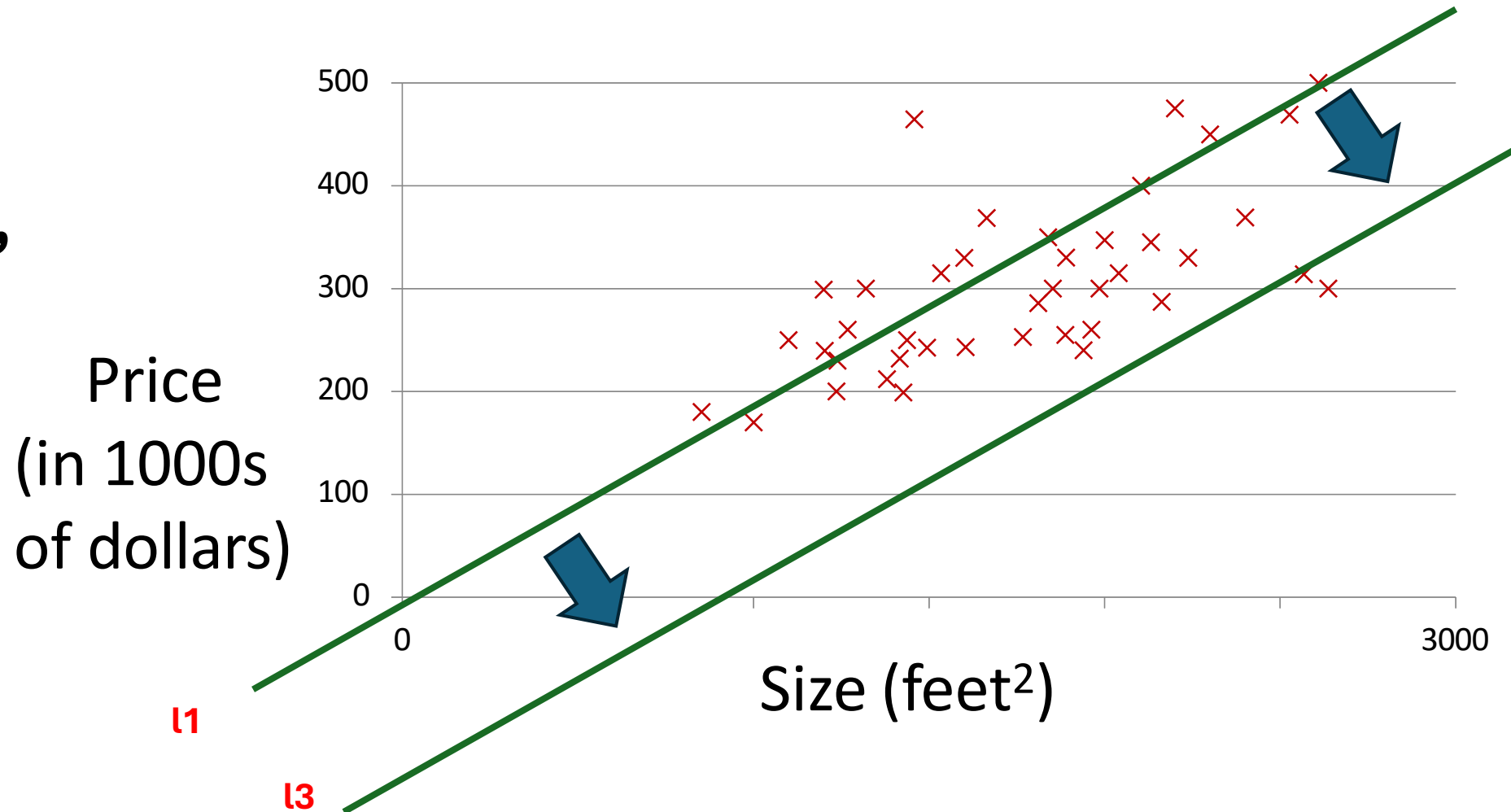
Regression- keeping slope b_1 constant

**Housing
Prices
(Portland,
OR)**



Regression- keeping slope b_1 constant

**Housing
Prices
(Portland,
OR)**



Regression- keeping slope b_1 constant

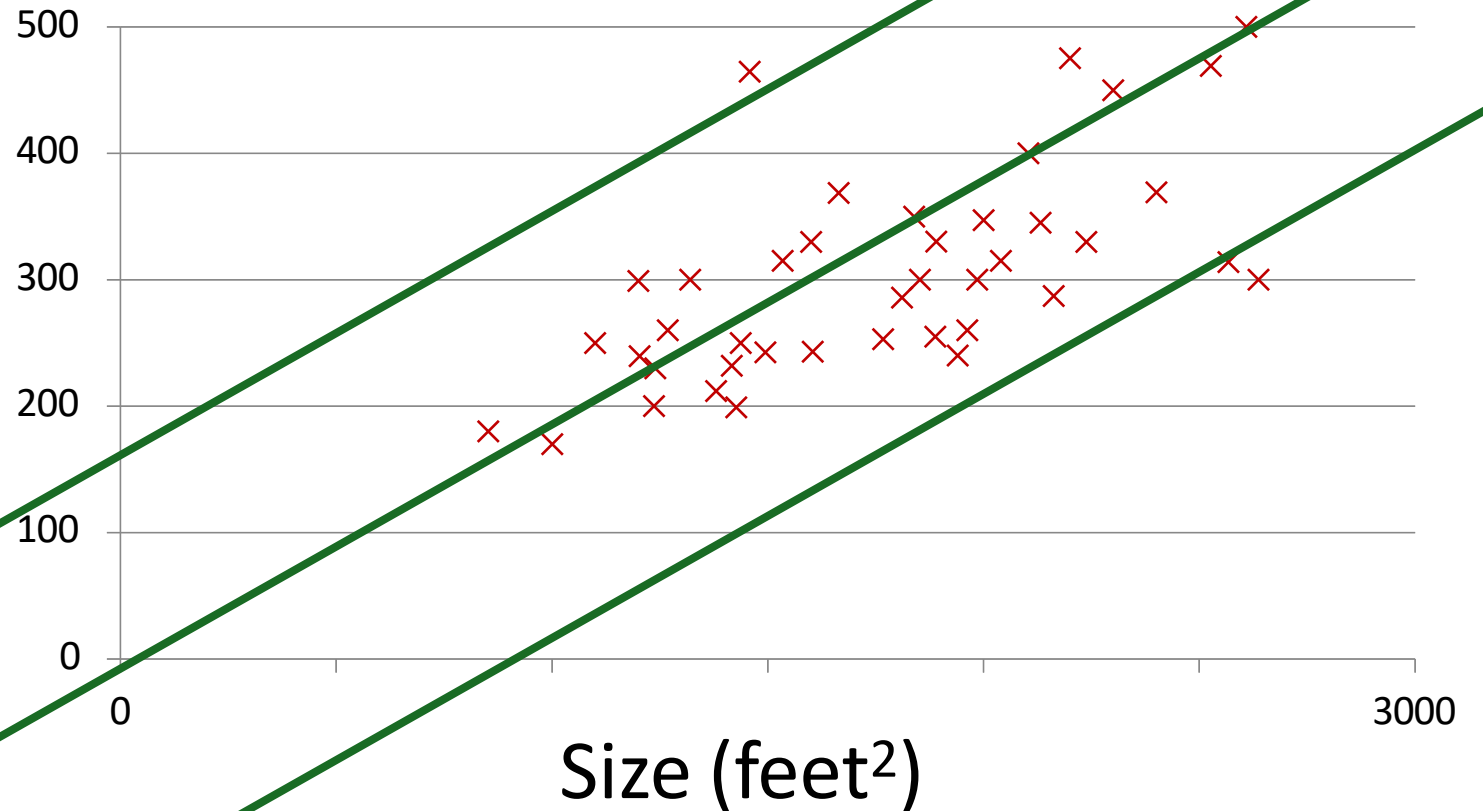
**Housing
Prices
(Portland,
OR)**

Price
(in 1000s
of dollars)

l_2

l_1

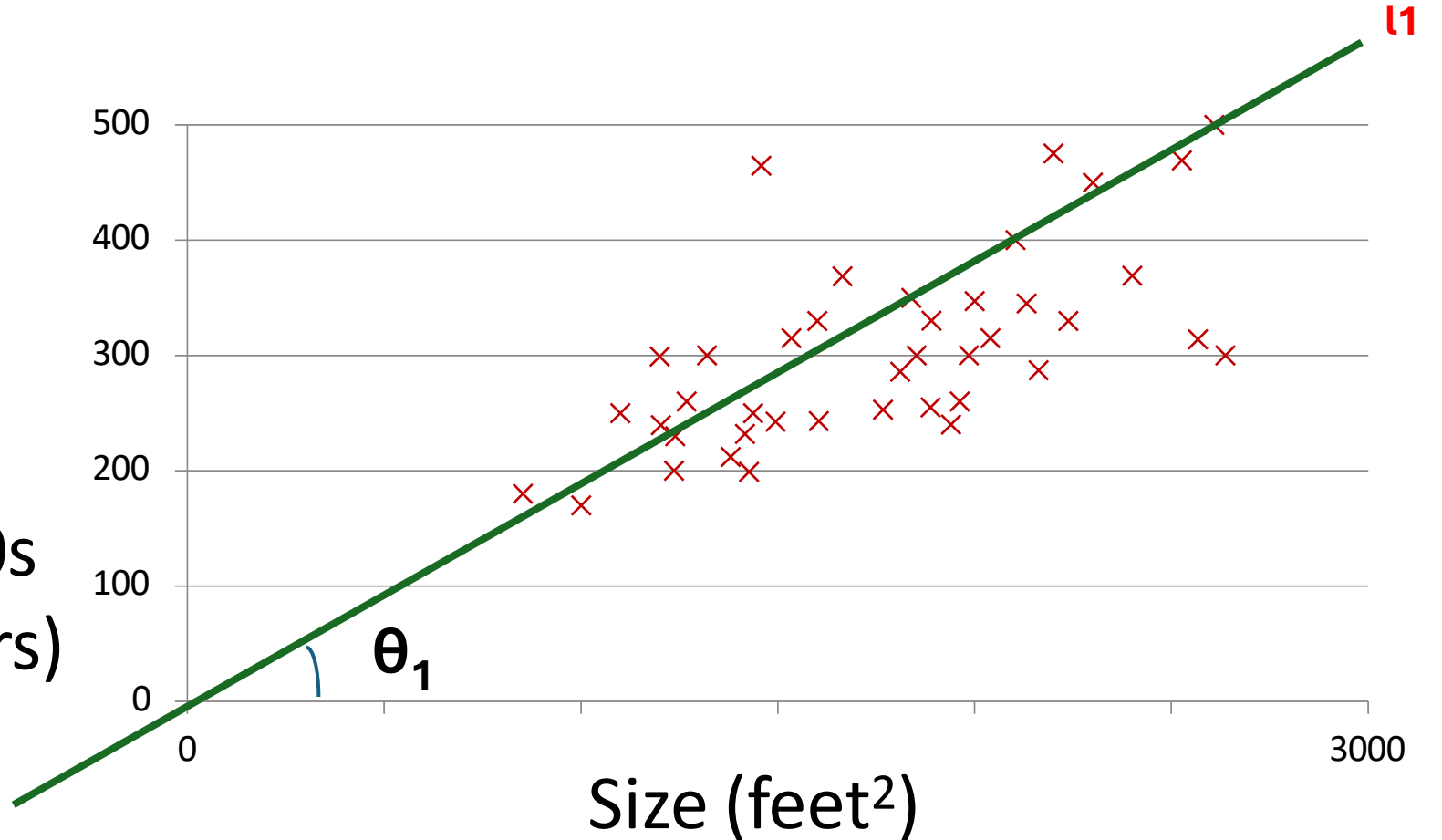
l_3



Regression

Housing Prices (Portland, OR)

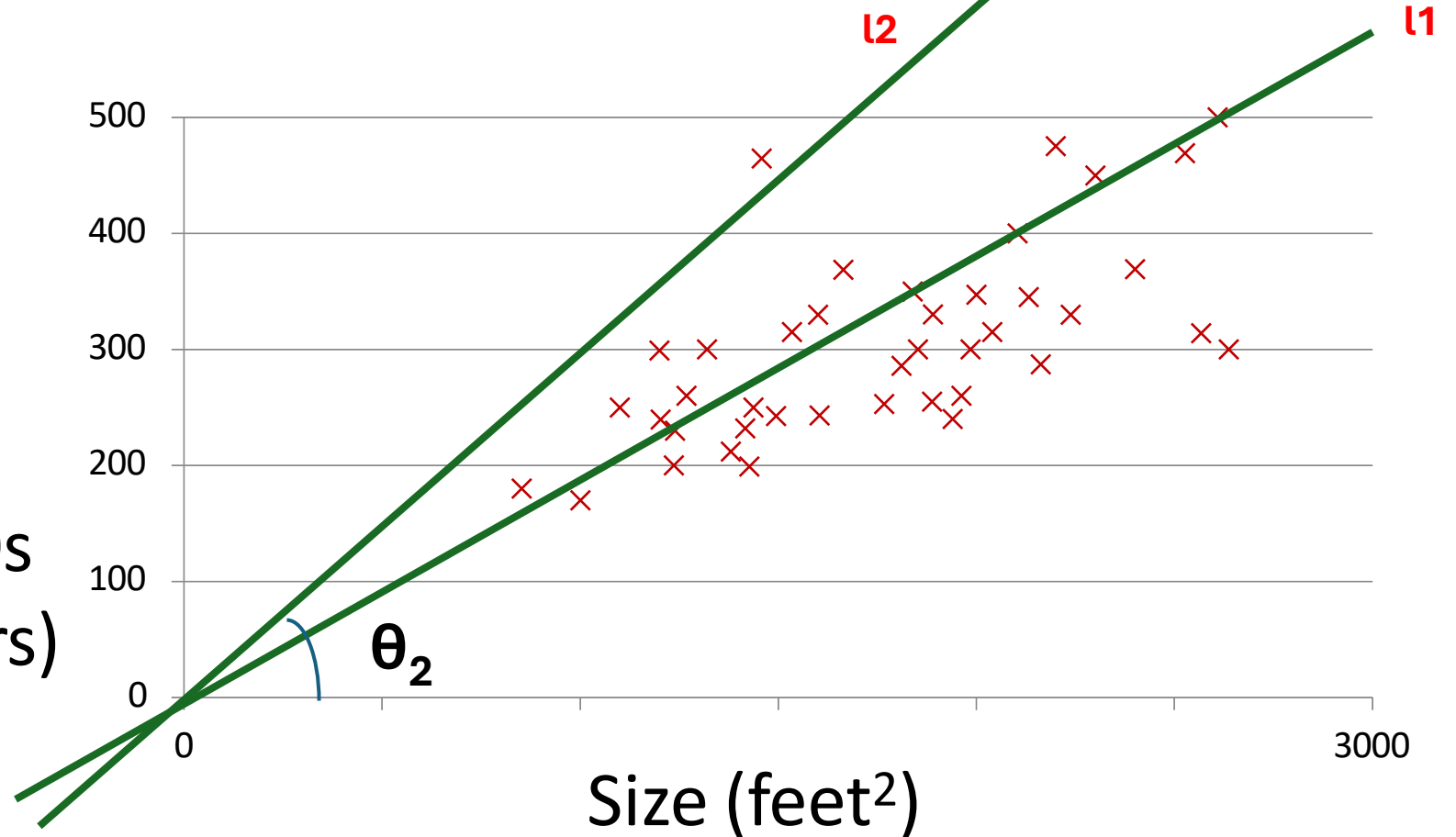
Price
(in 1000s
of dollars)



Regression- keeping y-intercept b_0 constant

Housing Prices (Portland, OR)

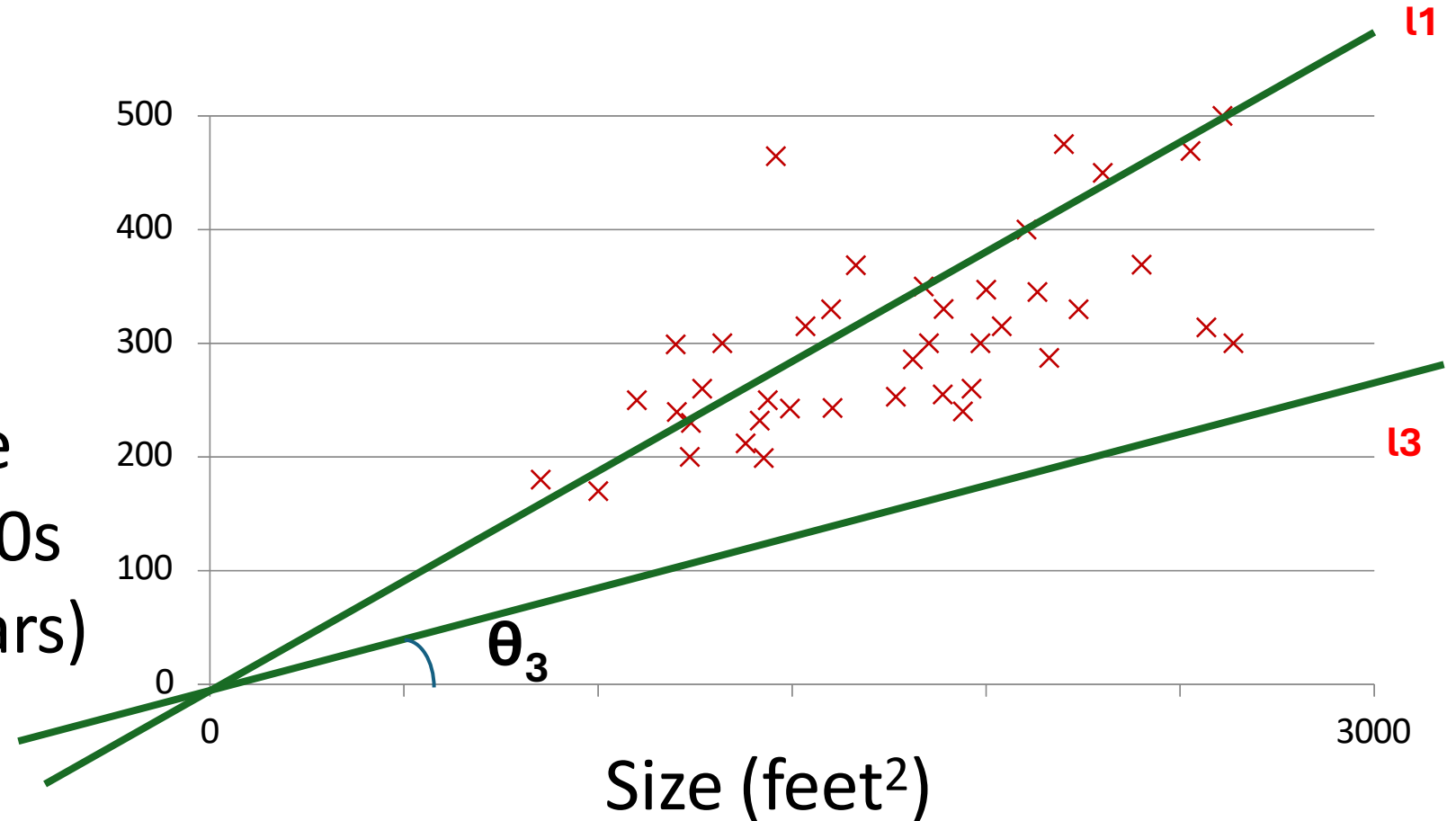
Price
(in 1000s
of dollars)



Regression- keeping y-intercept b_0 constant

Housing Prices (Portland, OR)

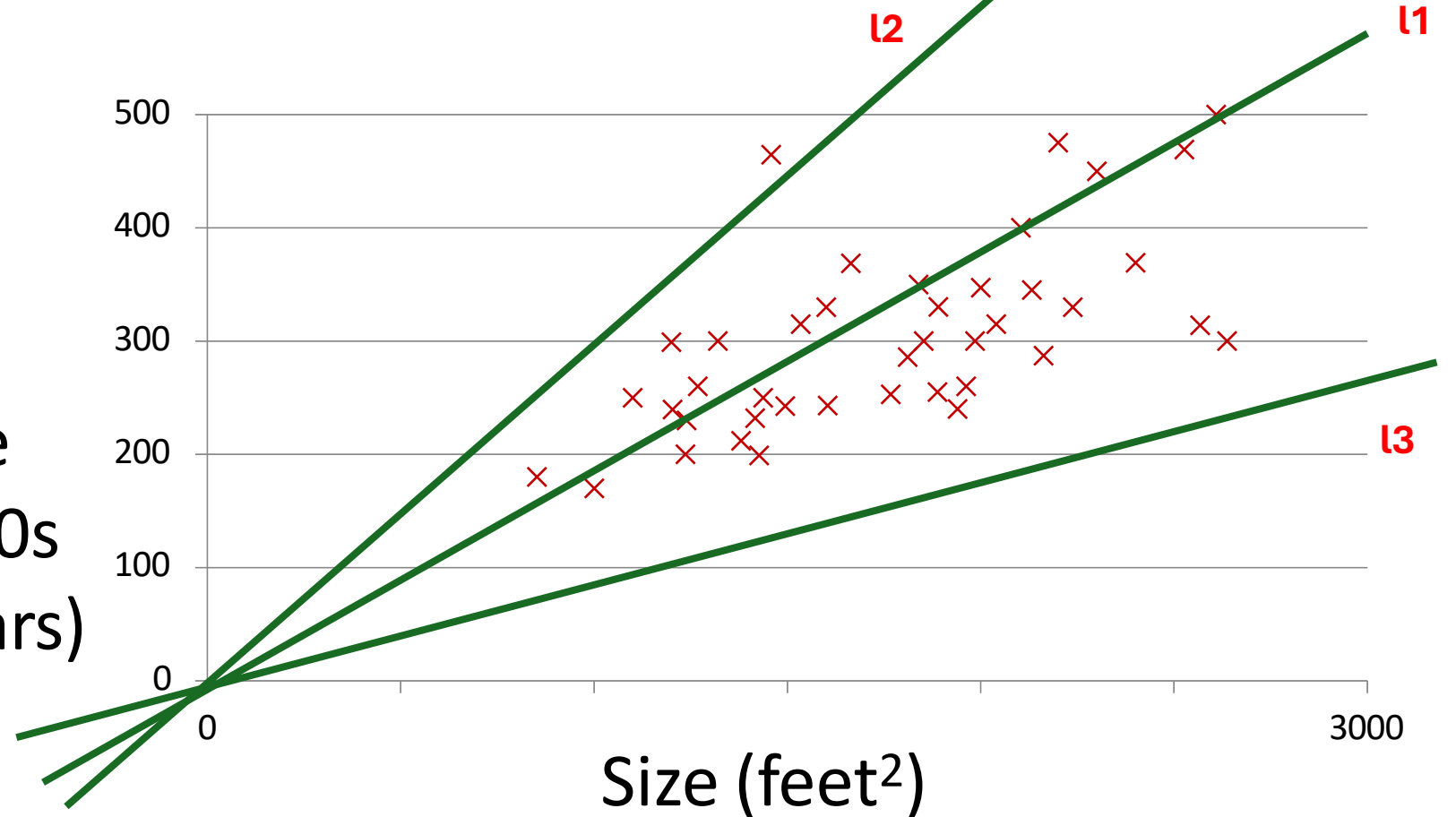
Price
(in 1000s
of dollars)



Regression- keeping y-intercept b_0 constant

**Housing
Prices
(Portland,
OR)**

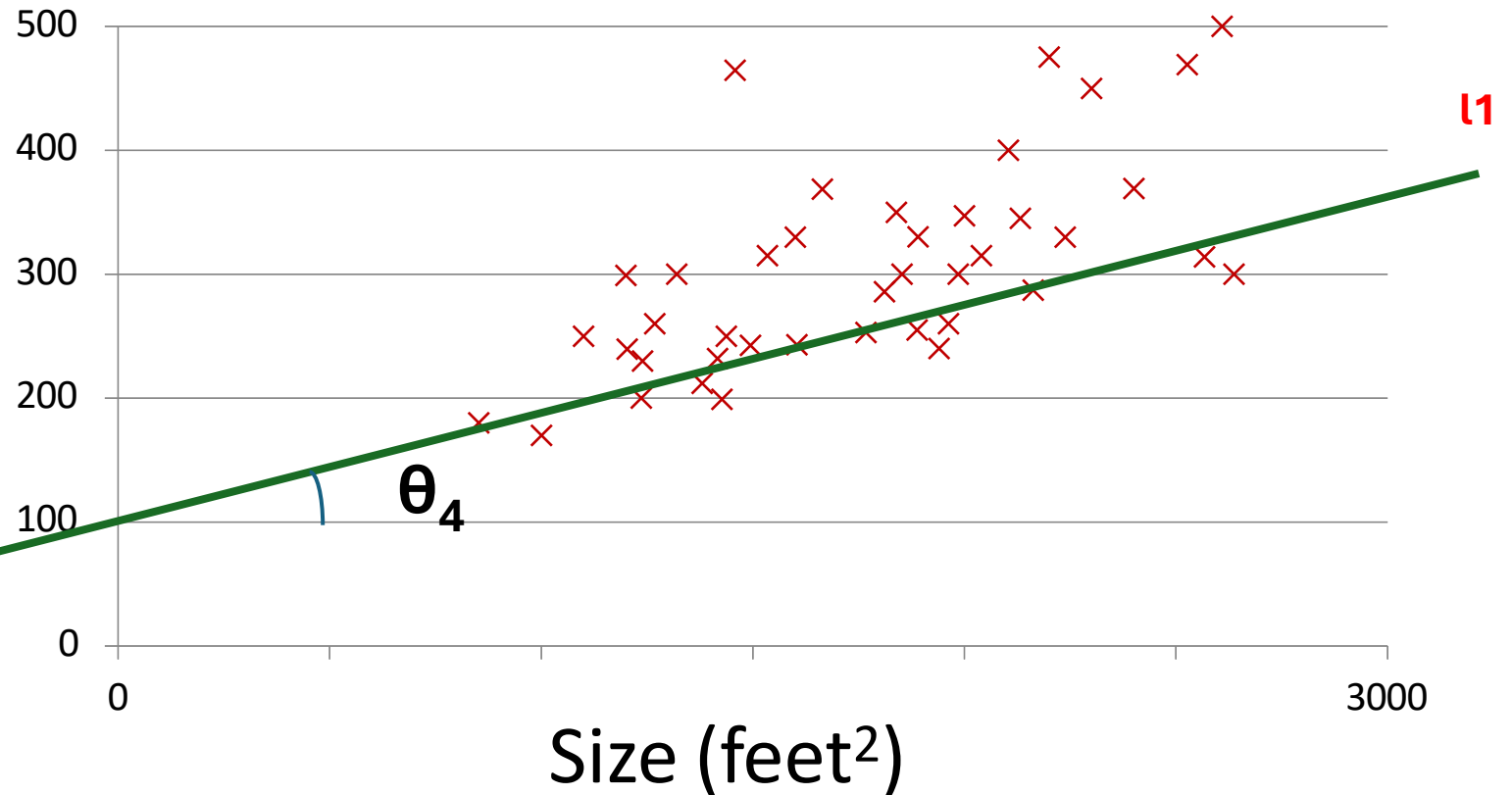
Price
(in 1000s
of dollars)



Regression- changing b_0 and b_1

Housing
Prices
(Portland,
OR)

Price
(in 1000s
of dollars)

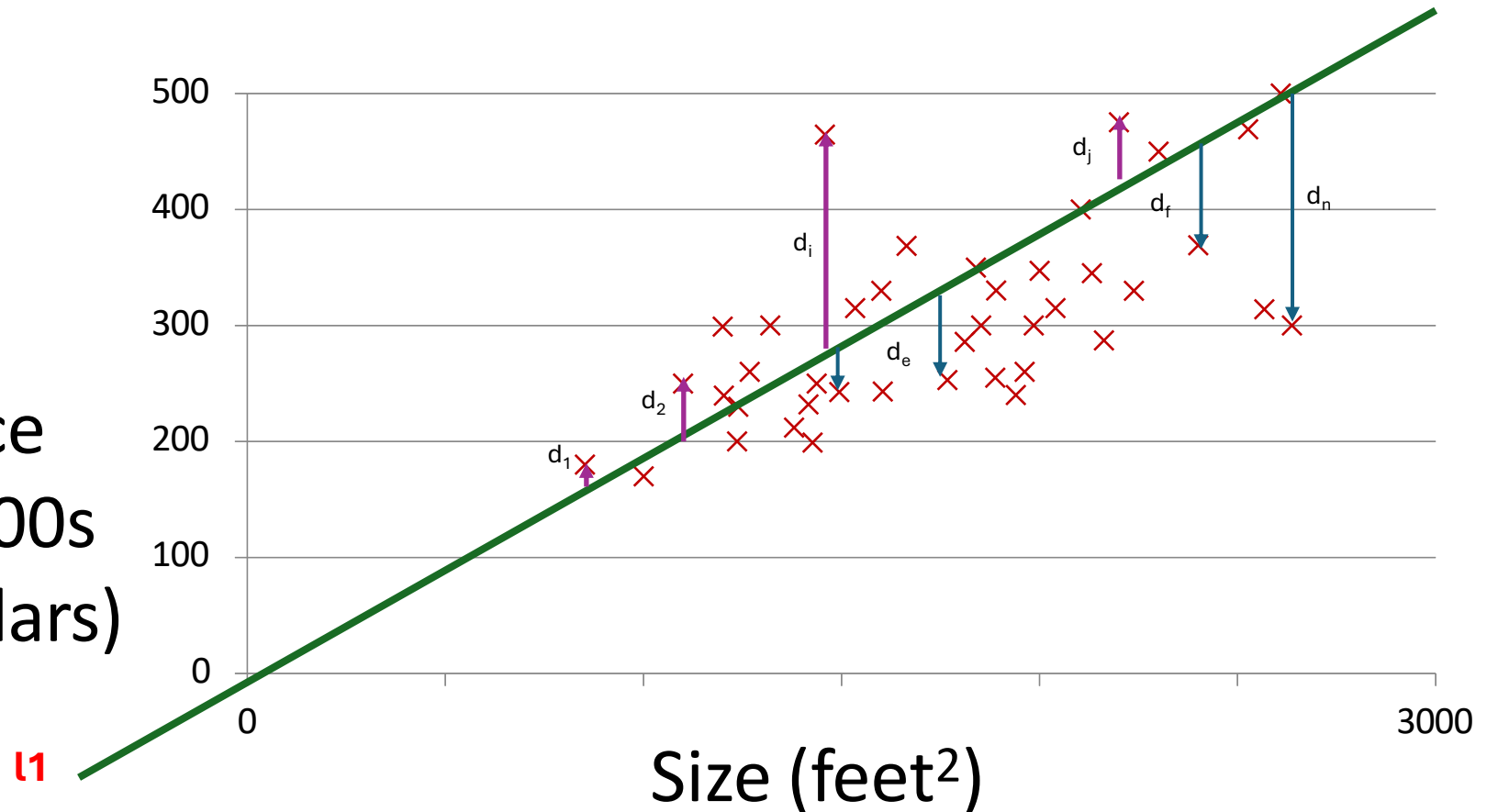


Regression

$$\text{Error} = \epsilon = d_1^2 + d_2^2 + \dots + d_n^2$$

Housing Prices (Portland, OR)

Price
(in 1000s
of dollars)



Loss function- Mean Square Error (MSE)

- A loss function measures how far a model's prediction is from the actual (true) value.

$$\text{MSE} = \sum (y_i - \hat{y}_i)^2 / n$$

- where:

y_i = observed value of the dependent variable for the i th observation

$\hat{y}_i$ = estimated value of the dependent variable for the i th observation

Least Squares Method

- Least Squares Criterion

$$\min \sum (y_i - \hat{y}_i)^2$$

- where:

y_i = observed value of the dependent variable for the i th observation

$\hat{y}_i$ = estimated value of the dependent variable for the i th observation

Least Squares Method

- A Slope for the Estimated Regression Equation

- where:
$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

x_i = value of independent variable for i th observation

y_i = value of dependent variable for i th observation

$\bar{x}$ = mean value for independent variable

$\bar{y}$ = mean value for dependent variable

Least Squares Method

- y-Intercept for the Estimated Regression Equation

$$b_0 = \bar{y} - b_1 \bar{x}$$

Simple Linear Regression Equation

- Reed Auto periodically has a special week-long sale. As part of the advertising campaign Reed runs one or more television commercials during the weekend preceding the sale. Data from a sample of 5 previous sales are shown on the next slide.

<u>Number of TV Ads (x)</u>	<u>Number of Cars Sold (y)</u>
1	14
3	24
2	18
1	17
3	27
<u>$\Sigma x = 10$</u> $\bar{x} = 2$	<u>$\Sigma y = 100$</u> $\bar{y} = 20$

Solve it on board

$$b_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

Simple Linear Regression Equation

Sr.	Number of TV Ads (x)	Number of Cars Sold (y)	(x-x')	(y- $\bar{y}$)	(x-x') ²	(x-x')*(y- $\bar{y}$)
1	1	14				
2	3	24				
3	2	18				
4	1	17				
5	3	27				
Sum	$\sum x = ?$	$\sum y = ?$			$\sum (x-x')^2$	$\sum (x-x')*(y-\bar{y})$
Mean	$\bar{x}' = ?$	$\bar{y} = ?$			= ?	= ?

Simple Linear Regression Equation

Sr.	Number of TV Ads (x)	Number of Cars Sold (y)	(x-x')	(y- $\bar{y}$)	(x-x') ²	(x-x')*(y- $\bar{y}$)
1	1	14	-1	-6	1	6
2	3	24	1	4	1	4
3	2	18	0	-2	0	0
4	1	17	-1	-3	1	3
5	3	27	1	7	1	7
Sum	$\sum x = 10$	$\sum y = 100$			$\sum (x-x')^2$	$\sum (x-x')*(y-\bar{y})$
Mean	$x' = 2$	$\bar{y} = 20$			= 4	= 20

Simple Linear Regression Equation

- Slope for the Estimated Regression Equation

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{20}{4} = 5$$

- y -Intercept for the Estimated Regression Equation

$$b_0 = \bar{y} - b_1\bar{x} = 20 - 5(2) = 10$$

- Estimated Regression Equation

$$\hat{y} = 10 + 5x$$

Variability

- Variability refers to the degree to which data values differ from each other or from a central value (usually the mean).
- It quantifies how spread out the data are.
 - Low variability → Data points are close together
 - High variability → Data points are widely spread

Variability

- Variability measures dispersion around the mean:

$$\text{Deviation} = (X_i - \bar{X})$$

- Common measures of variability:

- Range

$$\max(X) - \min(X)$$

- Variance

$$\sigma^2 = \frac{\sum (X_i - \bar{X})^2}{n}$$

- Standard Deviation

$$\sigma = \sqrt{\text{Variance}}$$

Coefficient of Determination

- Relationship Among SST, SSR, SSE

$$\begin{array}{c} \text{SST} \quad = \quad \text{SSR} \quad + \quad \text{SSE} \\ \swarrow \quad \downarrow \quad \searrow \\ \sum (y_i - \bar{y})^2 = \sum (\hat{y}_i - \bar{y})^2 + \sum (y_i - \hat{y}_i)^2 \end{array}$$

where:

SST = total sum of squares

SSR = sum of squares due to regression

SSE = sum of squares due to error

Coefficient of Determination

Sr.	Number of TV Ads (x)	Number of Cars Sold (y)	$\hat{Y} = 10 + 5x$	SST $(y_i - \bar{y})^2$	SSR $(\hat{y}_i - \bar{y})^2$	SSE $(y_i - \hat{y}_i)^2$
1	1	14				
2	3	24				
3	2	18				
4	1	17				
5	3	27				
Sum	$\sum x = ?$	$\sum y = ?$		$\sum (y_i - \bar{y})^2 = ?$	$\sum (y_i - \bar{y})^2 = ?$	$\sum (y_i - \hat{y}_i)^2 = ?$
Mean	$\bar{x} = ?$	$\bar{y} = ?$				

Coefficient of Determination

Sr.	Number of TV Ads (x)	Number of Cars Sold (y)	$\hat{Y} = 10 + 5x$	SST $(y_i - \bar{y})^2$	SSR $(\hat{y}_i - \bar{y})^2$	SSE $(y_i - \hat{y}_i)^2$
1	1	14	15	36	25	1
2	3	24	25	16	25	1
3	2	18	20	4	0	4
4	1	17	15	9	25	4
5	3	27	25	49	25	4
Sum	$\sum x = 10$	$\sum y = 100$		$\sum (y_i - \bar{y})^2 = 114$	$\sum (y_i - \bar{y})^2 = 100$	$\sum (y_i - \hat{y}_i)^2 = 14$
Mean	$\bar{x} = 2$	$\bar{y} = 20$				

Coefficient of Determination

- It tells how well the regression model explains the variability of the dependent variable.
- The coefficient of determination is:

$$r^2 = SSR/SST = 1 - SSE/SST$$

where:

SST = total sum of squares

SSR = sum of squares due to regression

SSE = sum of squares due to error

Coefficient of Determination

- $r^2=0 \rightarrow$ Model explains none of the variability
- $r^2 =1 \rightarrow$ Perfect fit
- $r^2=0.75 \rightarrow$ 75% of variation in y is explained by x

Coefficient of Determination

- **Reed Auto Example:**

$$r^2 = SSR/SST = 100/114 = .8772$$

- The regression relationship is very strong; 87.72% of the variability in the number of cars sold can be explained by the linear relationship between the number of TV ads and the number of cars sold.

Sample Correlation Coefficient

- The sample correlation coefficient quantifies the strength and direction of the linear association between two variables.

$$r_{xy} = (\text{sign of } b_1) \sqrt{\text{Coefficient of Determination}}$$

$$r_{xy} = (\text{sign of } b_1) \sqrt{r^2}$$

- where:

b_1 = the slope of the estimated regression equation

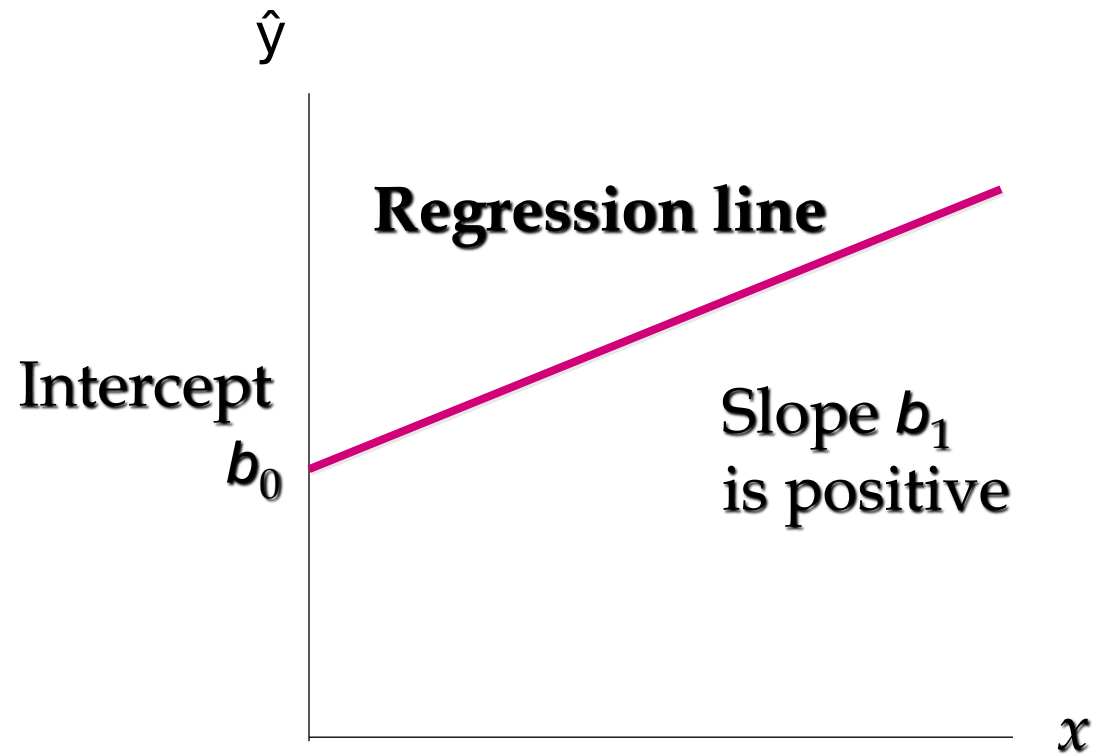
Sample Correlation Coefficient

- $-1 \leq r_{xy} \leq 1$
- $r_{xy} > 0 \rightarrow$ **Positive linear relationship**
 - when one variable increases, the other also increases
- $r_{xy} < 0 \rightarrow$ **Negative linear relationship**
 - when one increases, the other decreases
- $r_{xy} = 0 \rightarrow$ **No linear relationship**
- $r_{xy} = 1 / -1 \rightarrow$ **Strong relationship**

- **Sign** $\rightarrow$ Direction (positive/negative)
- **Magnitude** $\rightarrow$ Strength of linear association
- Measures linear relationship only (not nonlinear)

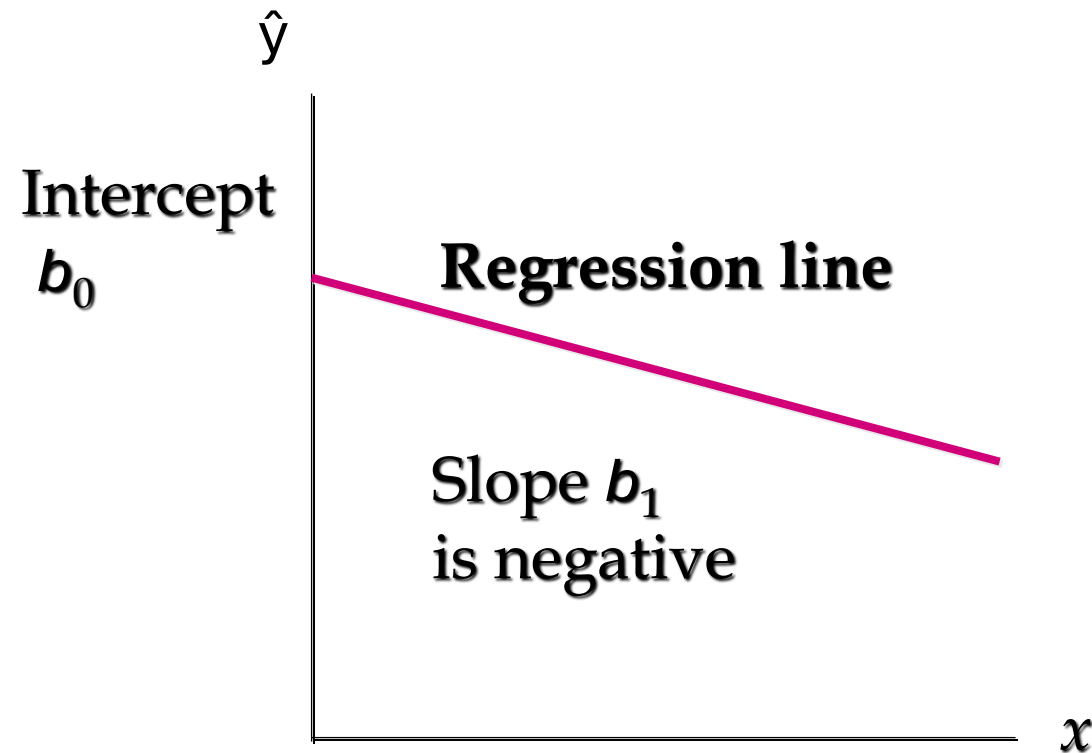
Simple Linear Regression Equation

- Positive Linear Relationship



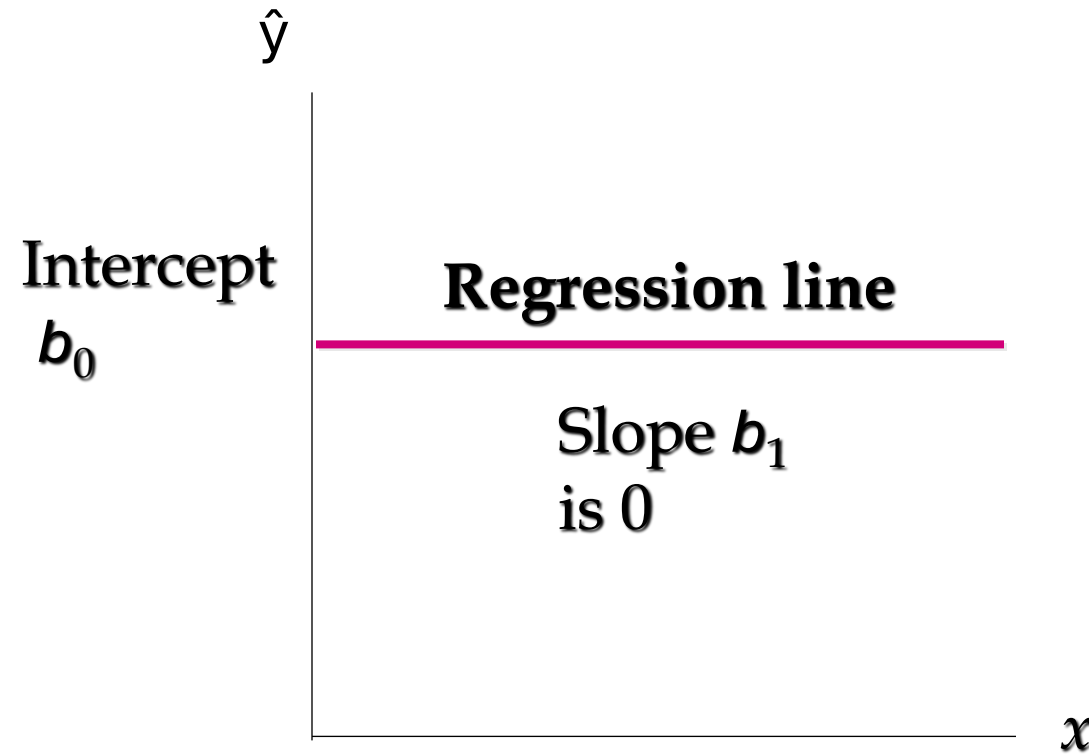
Simple Linear Regression Equation

- Negative Linear Relationship



Simple Linear Regression Equation

- No relationship



Sample Correlation Coefficient

$$r_{xy} = (\text{sign of } b_1) \sqrt{r^2}$$

- The sign of b_1 in the equation $\hat{y} = 10 + 5x$ is “+”.

$$r_{xy} = +\sqrt{.8772}$$

$$r_{xy} = +.9366$$

References
